Weekly Research Journal

ROS 2 Cartographer IMU Integration and Sensor Validation

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F1TENTH Autonomous Vehicle Project

1 October 2026

Abstract

This week integrated the VESC IMU into the ROS 2 Cartographer pipeline, activated the required base_link-to-imu transform, and verified simultaneous LiDAR and IMU operation. The IMU published at approximately 50 Hz. An initially intermittent LiDAR stream of approximately 0.06-0.17 Hz, including gaps up to 24.432 s, was traced to two Hokuyo driver processes accessing the same sensor. Removing the duplicate restored /scan to approximately 10 Hz. Cartographer then subscribed to both sensors, produced a continuously updating map-to-base_link transform, and published /map at 1.000 Hz. This establishes the basic LiDAR-IMU SLAM data path; RViz alignment, controlled-motion validation, speed estimation, and quantitative LiDAR-only versus LiDAR-IMU comparison remain pending.

1 Weekly contribution

The main contribution was not merely launching Cartographer, but isolating two independent configuration faults and documenting a reproducible sensor-to-SLAM interface. Table 1 separates verified results from measurements that were not yet performed.

Table 1: Evidence and scope of the 1 October results.

Contribution	Evidence	Scope
IMU integration	Active TF and approximately 50 Hz IMU stream	Completed
LiDAR fault isolation	Duplicate driver removed; /scan restored to approximately 10 Hz	Completed
Cartographer pipeline	Dual subscription, map-to-base_link pose, /map at 1 Hz	Completed
Motion and accuracy	No controlled motion or external ground truth	Not validated

2 System configuration and data flow

The F1TENTH platform used ROS 2 Humble, a Hokuyo 2D LiDAR, and the IMU exposed by the VESC driver. Static transforms linked both sensor frames to base_link. Cartographer used imu as the tracking frame while publishing the vehicle pose through base_link.

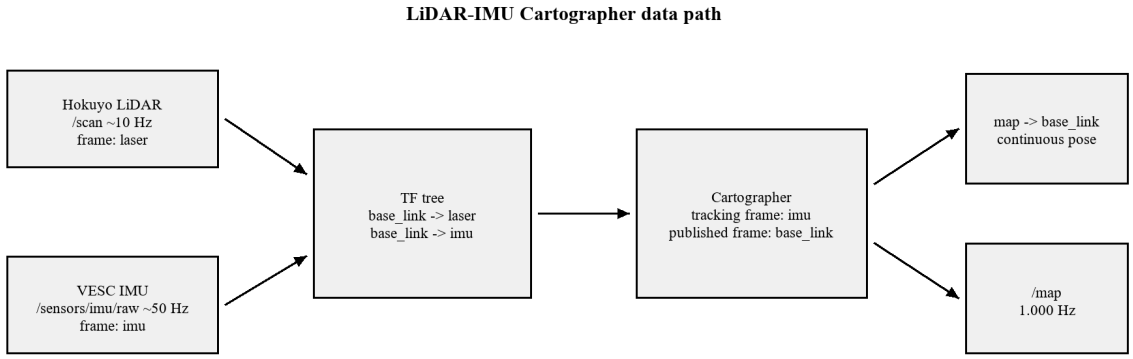


Figure 1: Verified LiDAR-IMU Cartographer data path. The sensor and map rates are measured observations; the transform arrows show the configured frame relationships.

Table 2: Verified interfaces and configuration.

Interface	Observed/configured value	Purpose
/scan	LaserScan; frame laser; approximately 10 Hz	Cartographer range input
/sensors/imu/raw	Imu; frame imu; approximately 50 Hz	Cartographer inertial input
base_link -> imu	Static transform enabled	Connect IMU to vehicle frame
tracking_frame	imu	Cartographer tracking frame
published_frame	base_link	Vehicle pose output frame
/map	Occupancy grid; 1.000 Hz	SLAM map output

2.1 Cartographer configuration changes

The Cartographer Lua configuration retained published_frame = base_link, changed tracking_frame to imu, and enabled TRAJECTORY_BUILDER_2D.use_imu_data. The launch file remapped scan to /scan and imu to /sensors/imu/raw. The robot bringup launch description was also corrected so the already-defined base_link-to-imu static transform was actually started.

3 Fault isolation and experimental validation

3.1 Duplicate Hokuyo driver diagnosis

The first LiDAR test was extremely slow and intermittent. Topic inspection reported two publishers, the ROS graph contained two nodes named /urg_node, and the operating system showed two urg_node_driver processes. One process had been started manually and the other by robot.launch.py. Stopping the manual instance restored the expected rate without any hardware change.

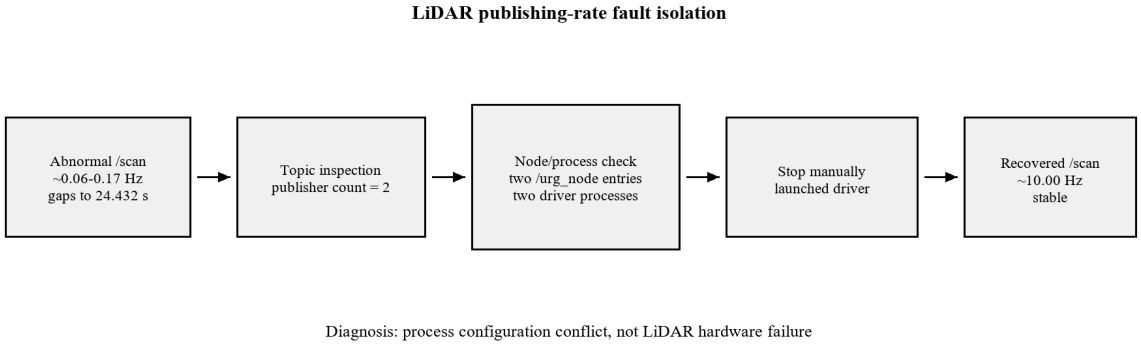


Figure 2: Evidence-based diagnostic sequence for the abnormal LiDAR rate. The recovered value shows that the failure originated in duplicated ROS processes rather than the physical sensor.

Table 3: Sensor and SLAM output measurements.

Measurement/condition	Observed result	Interpretation
VESC IMU	49.993, 50.001, 49.999 Hz	Stable
LiDAR with duplicate driver	approximately 0.06-0.17 Hz; gap to 24.432 s	Abnormal
LiDAR after duplicate removed	10.002, 10.005, 10.003 Hz	Stable
LiDAR interval after recovery	approximately 0.095-0.105 s	Consistent with 10 Hz
/map occupancy grid	1.000 Hz	Stable

3.2 Cartographer output verification

Node inspection confirmed Cartographer subscriptions to both /scan and /sensors/imu/raw. After startup, map-to-base_link became available and updated continuously. The /map topic was present at 1.000 Hz. These observations verify communication and transform availability, not trajectory accuracy or the quantitative benefit of the IMU.

4 Completed work and open work

Table 4: Project status at the end of the reporting week.

Status	Item
Completed	Enabled the base_link-to-imu static transform in the active launch description.
Completed	Verified stable IMU publication at approximately 50 Hz.
Completed	Identified and removed the duplicate Hokuyo driver process.
Completed	Restored /scan to approximately 10 Hz without changing hardware.
Completed	Configured Cartographer to use IMU data and verified both subscriptions.
Completed	Observed continuous map-to-base_link output and /map at 1 Hz.
Pending	Complete RViz map, LaserScan, and TF alignment checks.
Pending	Perform controlled translation and rotation tests.
Pending	Measure stationary drift and compare LiDAR-only with LiDAR-IMU operation.
Pending	Implement and validate a real-time vehicle-speed source.

5 Discussion and next experiment

The key engineering result is a stable multi-sensor SLAM input path. The LiDAR incident demonstrates that severe topic-rate degradation can arise from duplicated driver processes, while the TF incident shows that defining a transform node in source code is insufficient unless it is included in the returned launch description. Both faults were resolved through ROS graph and process-level evidence.

The next experiment should move the vehicle through controlled translations and rotations while recording Cartographer pose, map alignment, and IMU yaw behavior. A stationary segment should quantify pose drift. Only after those checks should LiDAR-only and LiDAR-IMU runs be compared, followed by a separate, calibrated investigation of vehicle speed.

Planned validation sequence

Step	Measurement	Decision enabled
1	RViz map, scan, and TF alignment	Verify geometric consistency
2	Stationary pose record	Measure drift and jitter
3	Controlled translation and rotation	Validate pose and yaw response
4	LiDAR-only versus LiDAR-IMU repeat	Assess IMU contribution
5	Independent speed reference	Begin calibrated velocity work